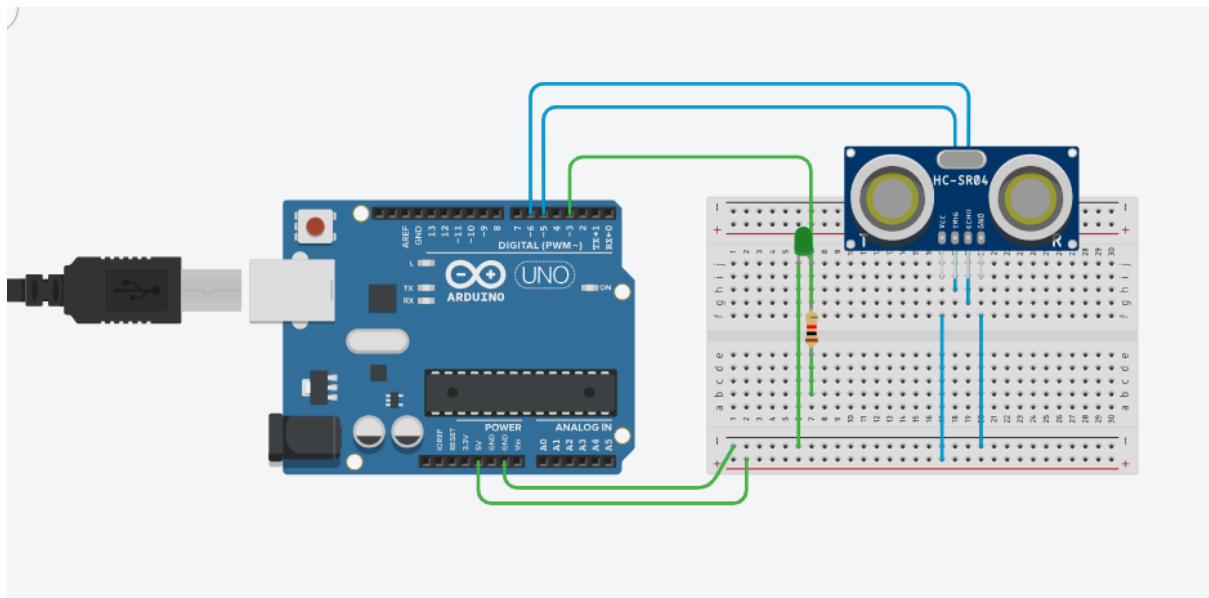


Robocon- Tinkercad Task-4.1

Circuit diagram and code explanation

Task-- ULTRASONIC SENSOR BASED LED BRIGHTNESS CONTROL

Circuit:



Explanation:

Components used—Arduino UNO, USD, breadboard, led, resistor, jumper wires

Code logic—

- We create a map function to convert the 10 bit input value to an 8 bit output value
- We use the usd mechanism- by sending a pulse through trig we observe the duration of echo being high then calculate distance using the duration
- For distance in range 20-200 led glows as per the map function or else stay in rest